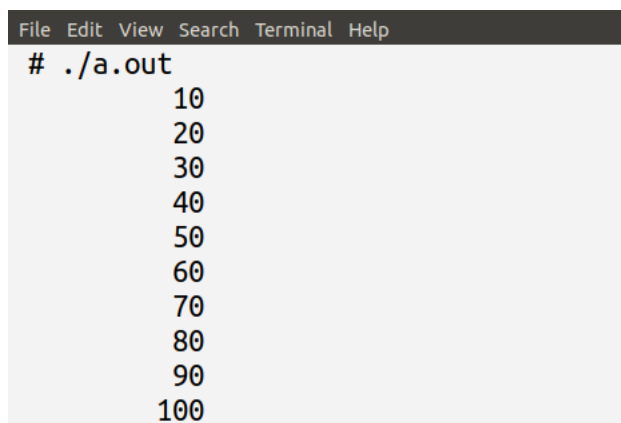


So the use of Fortran solutions is justified by their faster execution time when compared with solutions provided by scientific computing tools like Scilab. But if speed alone is the reason for using Fortran, why can't we switch to C or C++? Both these languages offer solutions that are extremely fast. But another fact in favour of Fortran is its relative ease of use. For example, array processing, very essential in scientific computing, is easier in Fortran when compared with C or C++. In Fortran, mathematics-like processing of arrays is possible. For example, every element of an array can be multiplied by a constant value without using a loop. Consider the Fortran program shown below:

```
program arr
  integer :: A(10), B(10)
  do i=1, 10
    A(i) = i
  end do
  B=A*10
  do i = 1, 10
    Print *, B(i)
  end do
end program arr
```

Figure 1 shows the output of this Fortran program. Notice that the statement 'B=A*10' to initialise array B would have been placed inside a loop in C or C++. Hence, because of faster execution and relative ease of use, Fortran is still widely used for scientific computing and high performance computing (HPC).

Finally, Fortran is still relevant because of the large amount of Fortran code still in use. In many fields of science, millions of lines of Fortran code are still in use and giving good performance. So, it is highly unlikely that all this programming code will be translated to code in some other programming language, that too without any real reason other than Fortran being not so popular among the general public.



```
# ./a.out
10
20
30
40
50
60
70
80
90
100
```

Figure 1: Array processing output in Fortran

So, what are the odds of a student learning programming for the first time in 2021 ever coming across a piece of code written in Fortran in his or her professional life? I don't want to tell any tall stories. The chances of young programmers encountering Fortran code is low unless they are working in the field of scientific computing or high performance computing. But since such computing is on the rise, the use of programming languages like Fortran may also rise in the future.

Now that we are convinced about the relevance of Fortran, let us learn a bit about its history and current status. Fortran was developed by John Backus (also famous for Backus-Naur Form of Context Free Grammars) and was operational in 1957. The language was initially called FORTRAN (abbreviation of Formula Translation). This also tells us that Fortran was always meant to be a scientific computing tool. There were a lot of standards for Fortran over its life time, with each revision bringing in new features. FORTRAN, FORTRAN II, FORTRAN III and FORTRAN IV were the first four noteworthy standards of Fortran. FORTRAN 66 and FORTRAN 77 were two standards of Fortran that were widely followed in the sixties and seventies. Notice that the name FORTRAN was changed to Fortran starting with Fortran 90, yet another very popular Fortran standard. Fortran 90 was very significant since it brought in many fresh changes to the language like



Figure 2: The logo of Fortran

allowing free-form source codes, recursive procedures, operator overloading, dynamic memory allocation, etc. Fortran 95, Fortran 2003 and Fortran 2008 are the later standards of Fortran, with Fortran 2018 being the latest. Now, Fortran is truly a general-purpose, compiled and imperative programming language. Though code written in older standards of Fortran like FORTRAN 77 is still surviving, the Fortran programming community today, generally, advises the use of Fortran 90 or a later standard. Such frequent upgradation of Fortran standards also tells us that the language is very much alive and evolving as quickly as any trending programming language. Figure 2 shows the logo of Fortran.

Next comes the important question. How do we execute a Fortran program? An open source compiler called gfortran, part of the GNU Compiler Collections (GCC), allows us to compile and execute Fortran programs in Linux. The gfortran compiler can be installed in Ubuntu Linux with the command *sudo apt install gfortran*. In other flavours of Linux also the installation of gfortran is very simple and requires the execution of just one command.